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MILITARY SPECIFICATION

FILTERS, LUBRICATING OIL, ENGINE, (LARGE)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 **Scope.**—This specification covers filters for use with internal combustion engines.

1.2 **Classification.**—Lubricating oil filters shall be of the following types, as specified (see 6.1):

Type I—By-pass.

Type II—Shunt.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-601 —Boxes, Wood-Cleated-Plywood, for Domestic Shipment.

NN-B-621 —Boxes, Wood, Nailed and Lock-Corner.

NN-B-631 —Boxes, Wood, Wirebound, for Domestic Shipment.

UU-C-843 —Cushioning Materials, Cellulosic.

MILITARY

JAN-P-105 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN-P-106 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.

JAN-P-116 —Packaging and Packing for Overseas Shipment—Preservation, Methods of.

JAN-B-121 —Barrier - Materials, Greaseproof.

JAN-P-127 —Packaging and Packing for Overseas Shipment—Tape, Adhesive, Pressure-Sensitive, Water Resistant.

JAN-P-132 —Packaging and Packing for Overseas Shipment—Crates; Unsheathed, Wood; Nailed (for Maximum Net Load of 2,500 Pounds).

JAN-N-514 —Name-Plate and Transportation-Data-Plate (for Construction and Engineering Equipment).

MIL-B-857 —Bolts, Nuts, Studs, and Tap Rivets (and Material for Same).

U. S. ARMY

3-182 —Compound, Insulation, Ignition.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

33G7 —Gaskets, Oil - Resisting (Sheet, Strip, and Molded).

66F4 —Filter-Elements, Lubricating Oil, Engine, Navy Standard (Large)

FSC 2945
2940

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STANDARDS

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MIL–STD–105 –Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking of Shipments.

DRAWINGS

BUREAU OF SHIPS

B-110 —Flanges, Steel, for Steel Tubing, Working Pressures of 150 Pounds for Sizes 10 Inches and Below, 50 Pounds for Sizes Above 10 Inches.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

NATIONAL BUREAU OF STANDARDS

Handbook H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

AMERICAN STANDARD ASSOCIATION

B16e—Steel Pipe Flange and Flange Fittings.

(Copies of the American Standard Association publications may be obtained upon application to the American Standard Association, 70th and East 45th Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 Design.

3.1.1 *General.*—The filter shall be of the multiple element design with removable filter elements, and shall consist of a filter case (body and cover or head). The design of the filter shall be such, when elements are mounted therein, as to permit the lubricating oil to enter the body, after which all or a part of the entering oil shall pass through the filter elements and the balance, if any, shall pass through one or more by-pass relief valves contained in the filter body. Dimensions shall be as shown on figures 1 to 4, inclusive, *unless otherwise specified by the bureau or agency concerned.*

3.1.2 *Filter case.*—Unless otherwise specified, the filter case shall be constructed of steel and shall be designed to use filter elements conforming to Specification 66F4. The clearance between elements, and between elements and the wall of the filter body, shall be at least $\frac{3}{8}$ inch when elements of the maximum allowable dimensions are used.

3.1.3 *Mounting of elements.*—Means shall be provided for mounting the filter elements in the filter body so as to be unaffected by vibration of engine or shock of gunfire, and to allow the elements to be removed from the body without disconnecting any of the attached piping. A hold-down device shall be provided for holding the elements securely in place. The device shall be so designed that oil will not leak by or through it from the inlet to the outlet side of the filter. The device shall be capable of exerting a longitudinal (compressive) force on each element contained not exceeding 500 pounds and not less than 100 pounds. The device shall be independent of the filter cover, and any nonmetallic gasket (see 3.1.5) or packing required in the device shall be so designed that it may be made readily from sheet packing with an ordinary gasket cutter.

3.1.4 *Filter covers.*—Filter covers shall be held in place by means of a minimum of eight swing-type hold-down bolts which shall be attached to the body. Means for lifting the cover, such as eye bolts or handles, shall be provided.

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3.1.5 *Gaskets.*—Gaskets shall be made of material conforming to Specification 33G7. Dimensions of the gasket between the filter cover and body shall be as shown on figures 1 to 4, inclusive. An annular ring or groove shall be incorporated in the body to insure installing the gasket properly and to prevent it from "blowing out".

3.1.6 *Standpipe.*—The standpipe shall be supported at the top by a spider or other suitable means, and shall be provided with an element stop located at the top level of the filter sump (see 3.1.10). The stop shall be attached to the standpipe by a leakproof joint. The dimensions of the standpipe and stop shall be as follows:

	Inches
Standpipe:	
Inside diameter, minimum	1½
Outside diameter, minimum	1.900
Outside diameter, maximum	2.010
Stop:	
Outside diameter, minimum	4
Thickness, minimum	⅛

3.1.7 *Ventcock.*—A ¼-inch iron pipe size ventcock shall be provided in the filter above the top surface of the top elements, at the highest point practicable.

3.1.8 *Gage connection.*—A ¼-inch iron pipe size tapped hole shall be provided in the body for gage connection.

3.1.9 *Bosses.*—Bosses 1¼ inches in outside diameter and ¾ inch high shall be provided for the gage connections and for the air vent connections.

3.1.10 *Sump.* — A sump not less than 2 inches deep and having a volume of not less than 50 cubic inches per element shall be provided below the elements for collection of water and sediment.

3.1.11 *Drain connection.*—The ¾-inch iron pipe size tapped drain connections shall be located so as to drain the lowest part of the clean and of the dirty oil compartments of the filter bodies.

3.1.12 *Bolts, nuts, and studs.*—Bolts, nuts and studs shall conform to Specification MIL–B–857.

3.1.13 *Threads.* — Threads shall conform to Handbook H28, American National Pipe Threads.

3.2 Hydrostatic pressure.—The filter case shall withstand a hydrostatic pressure of 150 p.s.i.

3.3 Finish.—The interior and exterior surfaces of the filter case shall be free from rust, scale, welding spatter, or other foreign matter. The exterior surfaces shall be suitably painted.

3.4 Drawings.—Before production, a detailed drawing of the filter assembly shall be furnished for the approval of the bureau or agency concerned.

3.5 Type I, by-pass filters.—Type I filters shall be as shown on figures 1 and 2.

3.5.1 *Orifice plate.*—A removable restrictive orifice plate shall be incorporated inside the filter body. Unless otherwise specified, the orifice size shall be as specified in table I.

TABLE I.—*Orifice size*

Number of elements	Orifice diameter
	Inch
3	0.200
4	.238
6	.281
8	.328

3.5.2 *Name-plates.* — A name-plate conforming to Specification JAN–N–514, showing the following data, shall be securely attached to each filter case:

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Manufacturer's name.
Filter type identification number.
Serial number.
"Standard (Large)".
Replacement element standard stock number.
Orifice diameter.

The Manufacturer's part number for the element shall not appear on the name-plate.

3.6 **Type II, shunt filters.**—Type II filters shall be as shown on figures 3 and 4.

3.6.1 *Baffle.*—A suitable baffle shall be provided to prevent the oil entering the filter from impinging on the filter elements.

3.6.2 *Name-plates.* — A name-plate conforming to Specification JAN–N–514 showing the following data shall be securely attached to each filter case:

Manufacturer's name.
Filter type identification number.
Serial number.
"Standard (Large)".
Replacement element standard stock number.

The manufacturer's part number for the element shall not appear on the name-plate.

3.7 **Flanges.**—Flanges shall be provided for the inlet and outlet connections (see figs. 1 to 4, incl.) and shall be plainly and permanently marked "inlet" and "outlet" as applicable. For the Army, flange dimensions shall be in conformance with ASA Standard B16e. For the Navy, flange dimensions shall be in accordance with type C or D of Drawing B–110. Size of connections shall be as specified in table II.

TABLE II.—*Size of connections*

Number of elements	Iron pipe size	
	Type I	Type II
8	1	2
4	1¼	2½
6	1¼	3
8	1½	4

3.8 **Valves.**—A relief (by-pass) valve or valves shall be located level with or above the top surface of the top elements, and shall be of sufficient capacity to handle the full capacity of the inlet and outlet flange connections at the maximum differential pressure specified in table III.

TABLE III.—*Maximum differential pressure*

	Type I	Type II
	p.s.i.	*p.s.i.*
Start open	25	12
Full open	30	17

Capacities of the inlet and outlet flanges and the element standpipe, based on a maximum velocity of 6 feet per second shall be as specified in table IV.

TABLE IV.—*Capacities of flanges*

Size	Capacity
Flange	Gallons per minute
1	17
1¼	30
1½	40
2	67
2½	102
3	156
4	265
Element standpipe: 1.5 inch inside diameter	33

The temperature of oil used to determine flow rate through valves shall be adjusted to produce an oil viscosity of 170 seconds Saybolt Universal during the test (see 4.5).

3.9 **Workmanship.** — The workmanship shall be first class in every respect.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. — For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Inspection lot.—For purposes of sampling and inspection, a lot shall consist of all filters of the same type and size offered for delivery at one time.

4.3 Sampling for lot acceptance.

4.3.1 *Sampling for inspection.*

4.3.1.1 *Sampling for intermittent procurement in isolated lots.*—A random sample of filters shall be selected from each lot of material delivered or offered for Government inspection of visual and dimensional characteristics and nameplates with lot acceptance based on the single sampling inspection specified in table V in accordance with Standard MIL–STD-105:

TABLE V.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

Number of filters in inspection lot	Number of filters in sample	Acceptance number (defectives)	Rejection number (defectives)
40 and under	10	0	1
41 to 110	15	0	1
111 to 800	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 to 3200	110	4	5
Over 3200	150	5	6

4.3.1.2 *Sampling for continuous production.*—Where there is continuous production for the Government and inspection lots include at least 200 items sampling for Government inspection shall be conducted in accordance with Standard MIL–STD–105 at Inspection Level II and AQL = 1.5 per cent defective, the type of sampling to be determined by the Government inspector.

4.3.2 *Sampling for tests.*—A random sample of filter cases shall be selected from each inspection lot and tested as specified in 4.5, with lot acceptance based on table VI in accordance with Standard MIL–STD–105:

TABLE VI.—*Sampling for hydrostatic tests*

Number of filter cases in inspection lot	Number of filter cases in sample	Acceptance number (defectives)	Rejection number (defectives)
40 and under	7	0	1
41 to 110	10	0	1
111 to 300	15	0	1
301 to 500	25	0	1
501 to 800	35	1	2
801 to 3200	50	2	3
Over 3200	75	3	4

4.4 Inspection.

4.4.1 *Visual and dimensional inspection.*—Each of the sample filters selected in accordance with the table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any filter in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective filters in any sample exceed the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5 Test procedures.

4.5.1 *Hydrostatic tests.*—Each of the sample filter cases selected in accordance with table VI shall be subjected to a hydrostatic pressure of 150 p.s.i. and the relief valves shall be tested for conformance with 3.8. Any filter case or valve failing this test shall be rejected, and if the number of failures in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6 Rejected lots.—A rejected lot may be resubmitted for Government inspection and

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tests only after the contractor has reworked or removed all nonconforming filters. The Government inspector shall again select samples from the resubmitted lots for inspection and/or testing to verify compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging for domestic or overseas shipment.

5.1.1 *Cleaning and drying.* — Before the application of any preservative, all surfaces to which a preservative is to be applied shall be cleaned and dried in conformance with Specification JAN-P-116, appendix I, using the most applicable methods of C-2 to C-13 described therein.

5.1.2 *Preservatives and application.*—Preservatives shall conform to the applicable specifications listed in table I of Specification JAN-P-116, for the types specified hereinafter. Application of preservatives shall be in conformance with Specification JAN-P-116, including appendix II, and as specified hereinafter. Preservatives shall be applied to unpainted surfaces but shall not be applied to painted surfaces or to gaskets.

5.1.3 *Repainting.* — When specified, (see 6.1), painted surfaces on which the paint film has become damaged shall have the damaged surfaces cleaned and repainted, the same color, in conformance with 3.3.

5.1.4 *Filter case.*—Interior surfaces of the steel filter case, including mountings for the elements, and the orifice plate, shall be coated with type P-10, grade I preservative of Specification JAN-P-116. Any excess preservative accumulating in the bottom of the case shall be drained. Exterior unpainted metal surfaces of the case, the inlet and outlet flanges, and all threaded surfaces, including bolts and nuts, shall be coated with type P-1 preservative of Specification JAN-P-116.

5.1.5 *Openings.* — The ventcock shall be closed and all openings into the filter case, including the ventcock, shall be covered with greaseproof barrier-material conforming to Specification JAN-B-121, type I, grade C, class 1, secured in place with pressure-sensitive tape conforming to Specification JAN-P-127, type I, grade B, color optional, of suitable width, coated with ignition insulation compound conforming to Specification 3-182.

5.2 Packing.

5.2.1 *For domestic shipment.*—The filters shall be packed in accordance with 5.2.1.1 or 5.2.1.2 as specified (see 6.1).

5.2.1.1 The filters shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, in such a manner as to insure acceptance and safe delivery to the designated point by common or other carriers, at the lowest rate of the carrier.

5.2.1.2 Filters, preserved as specified in 5.1 shall be packed in snug-fitting cleated plywood, nailed wood, or wirebound boxes conforming to Specification NN-B-601, NN-B-621, or NN-B-631, respectively, or in unsheathed wood crates conforming to Specification JAN-P-132. They shall be blocked against movement in the shipping container during handling and shipment.

5.2.2 *For overseas shipment.*—The filters preserved as specified in 5.1 shall be individually packed in nailed wood boxes conforming to Specification JAN-P-106, style 2, 2½ or 3, or wood cleated plywood boxes conforming to Specification JAN-P-105, style A or B for type 3 load. Suitable blocking and cushioning of creped cellulose wadding conforming to Specification UU-C-843, type III, class 3, shall be provided to prevent movement within the boxes. The boxes shall be closed and strapped in conformance with the applicable box specification.

5.3 Marking.—In addition to any special marking required by the contract or order,

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marking shall be in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of filter required (see 1.2).

(c) Whether repainting is required (see 5.1.3).

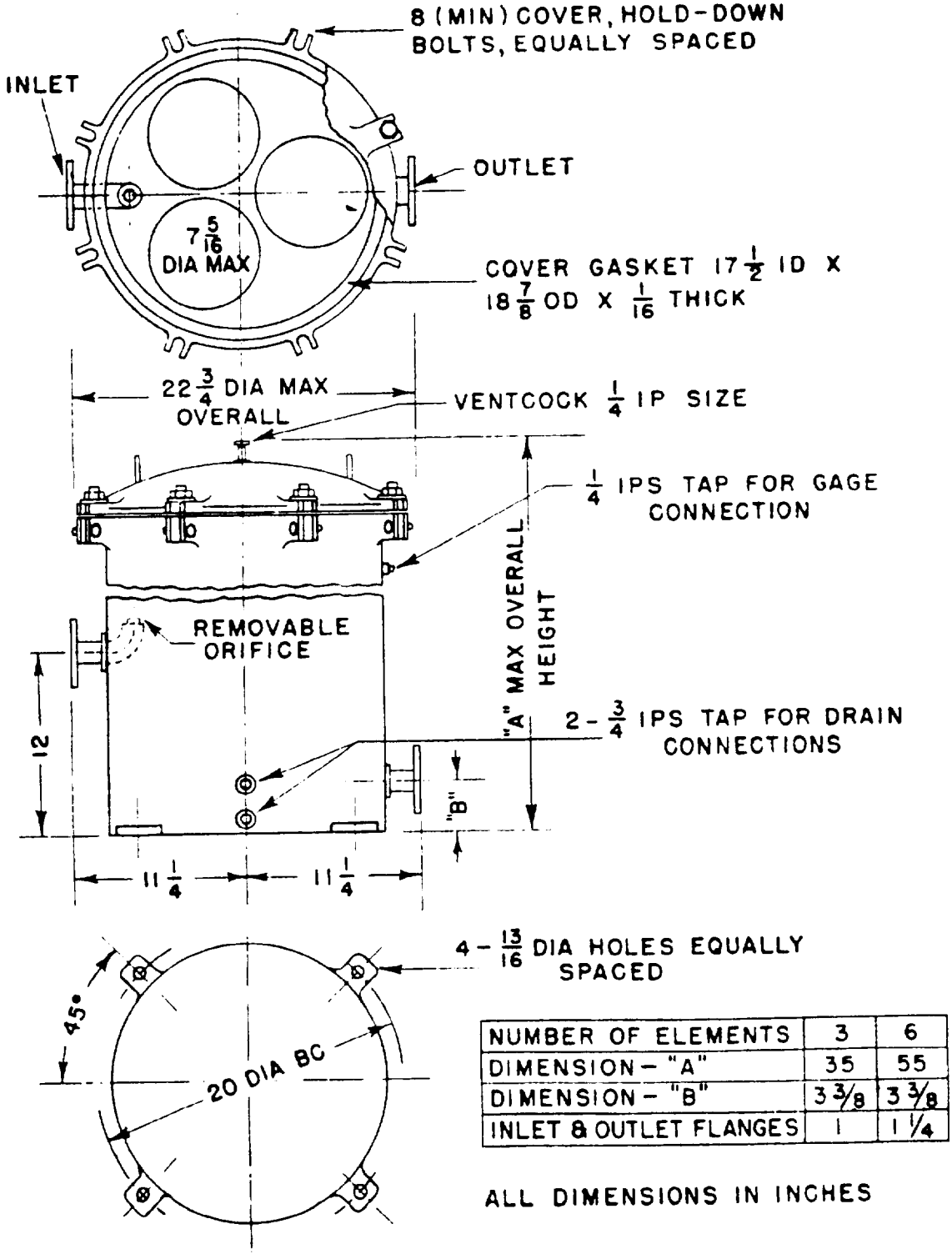
(d) Whether domestic or overseas packing is required, if domestic, the type required (see 5.2).

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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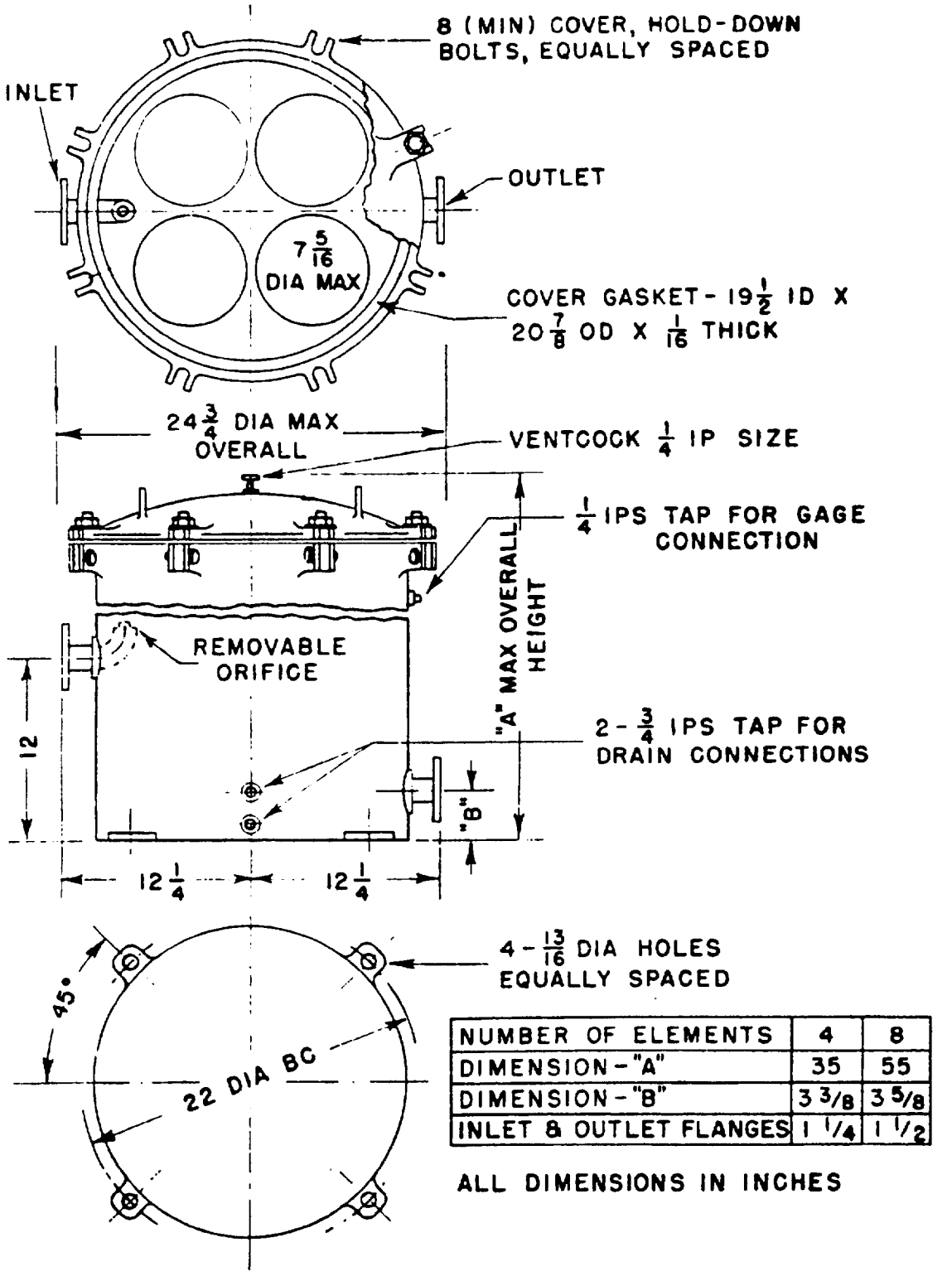


NUMBER OF ELEMENTS	3	6
DIMENSION - "A"	35	55
DIMENSION - "B"	3 3/8	3 3/8
INLET & OUTLET FLANGES	1	1 1/4

ALL DIMENSIONS IN INCHES

FIGURE 1.—*Type I standard lubricating oil filter cases for three and six large elements.*

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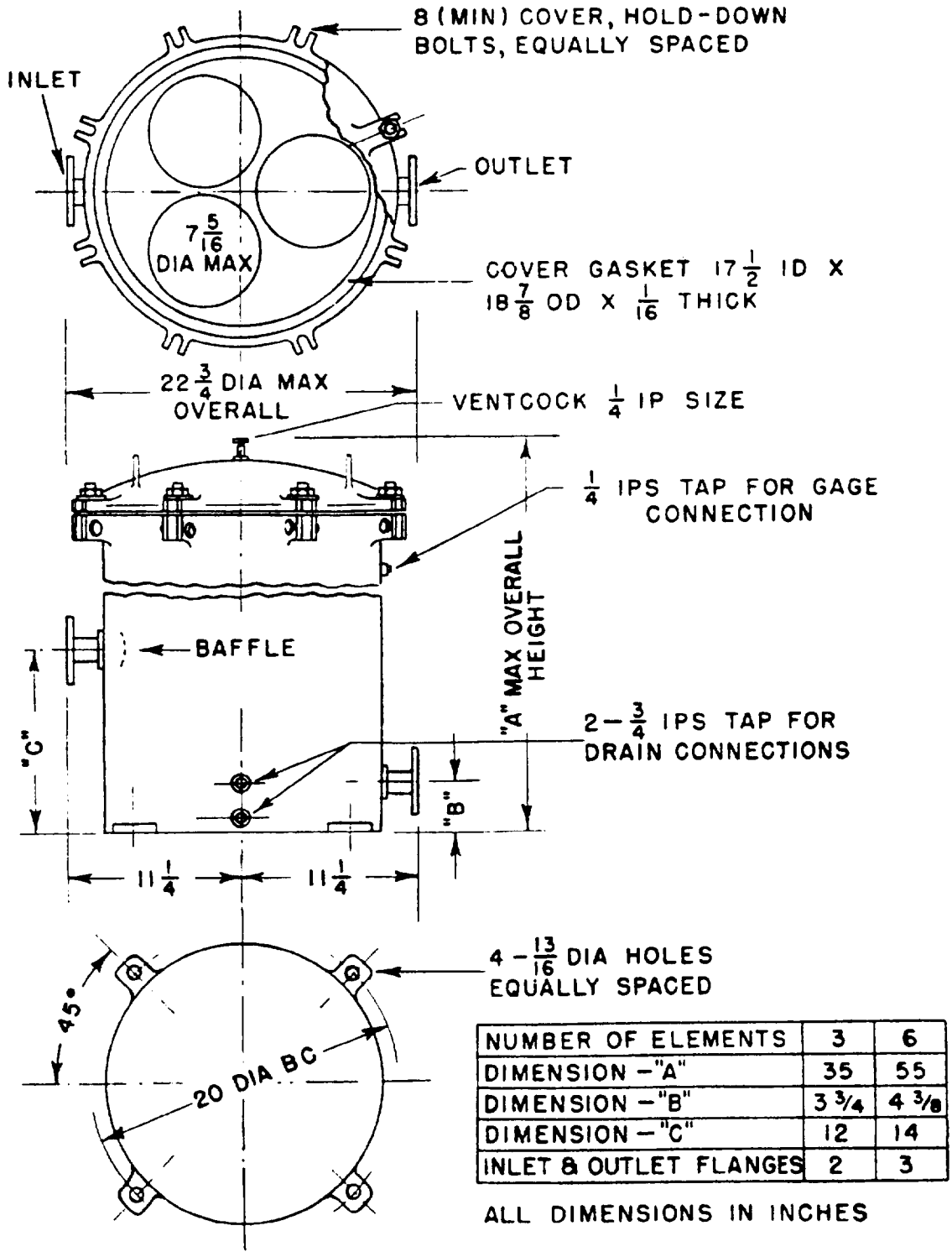


NUMBER OF ELEMENTS	4	8
DIMENSION - "A"	35	55
DIMENSION - "B"	3 3/8	3 5/8
INLET & OUTLET FLANGES	1 1/4	1 1/2

ALL DIMENSIONS IN INCHES

FIGURE 2.—*Type I standard lubricating oil filter cases for four and eight large elements.*

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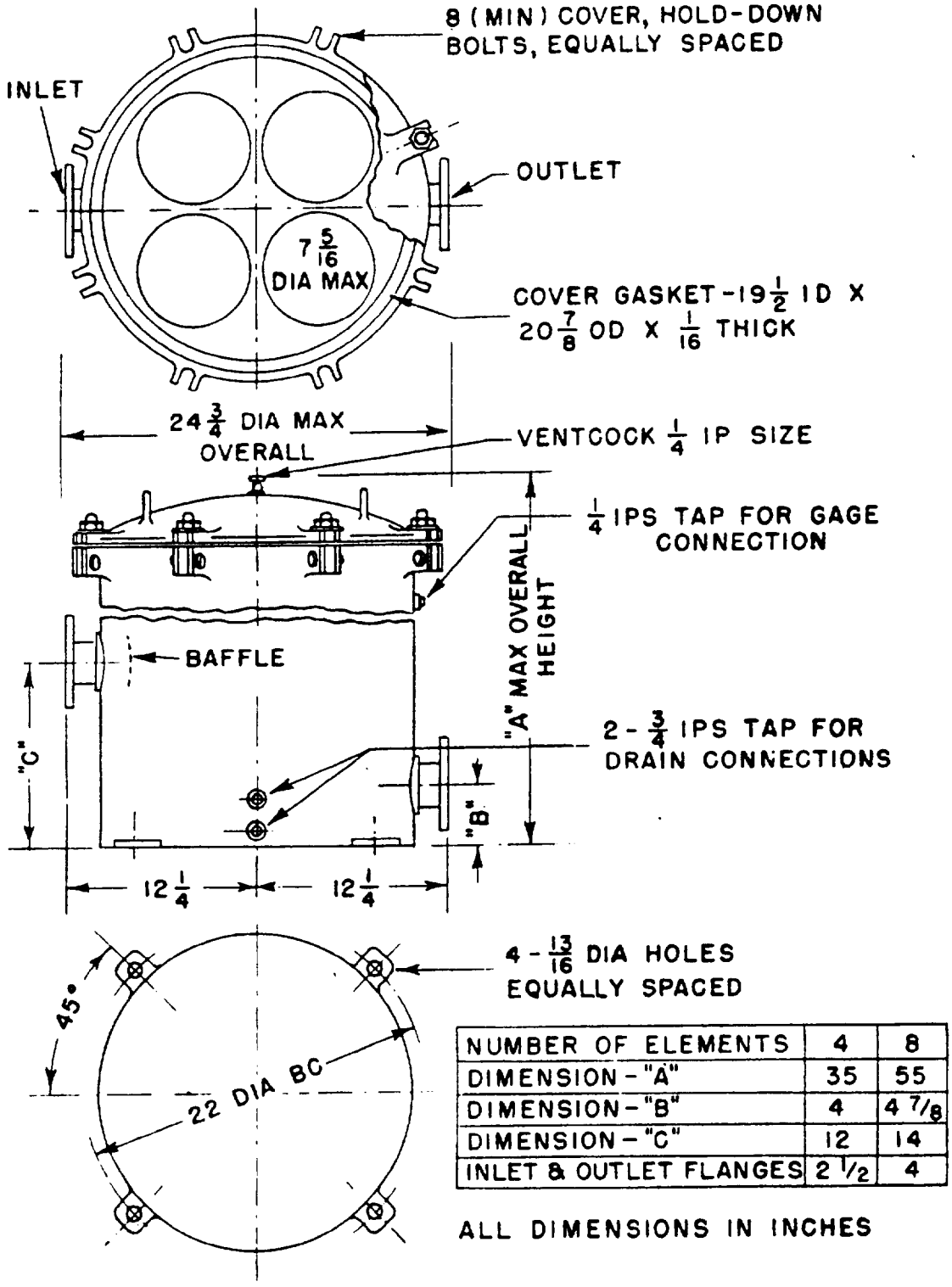


NUMBER OF ELEMENTS	3	6
DIMENSION — "A"	35	55
DIMENSION — "B"	3 3/4	4 3/8
DIMENSION — "C"	12	14
INLET & OUTLET FLANGES	2	3

ALL DIMENSIONS IN INCHES

FIGURE 3.—*Type II standard lubricating oil filter cases for three and six large elements.*

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NUMBER OF ELEMENTS	4	8
DIMENSION - "A"	35	55
DIMENSION - "B"	4	4 7/8
DIMENSION - "C"	12	14
INLET & OUTLET FLANGES	2 1/2	4

ALL DIMENSIONS IN INCHES

FIGURE 4.—*Type II standard lubricating oil filter cases for four and eight large elements.*

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